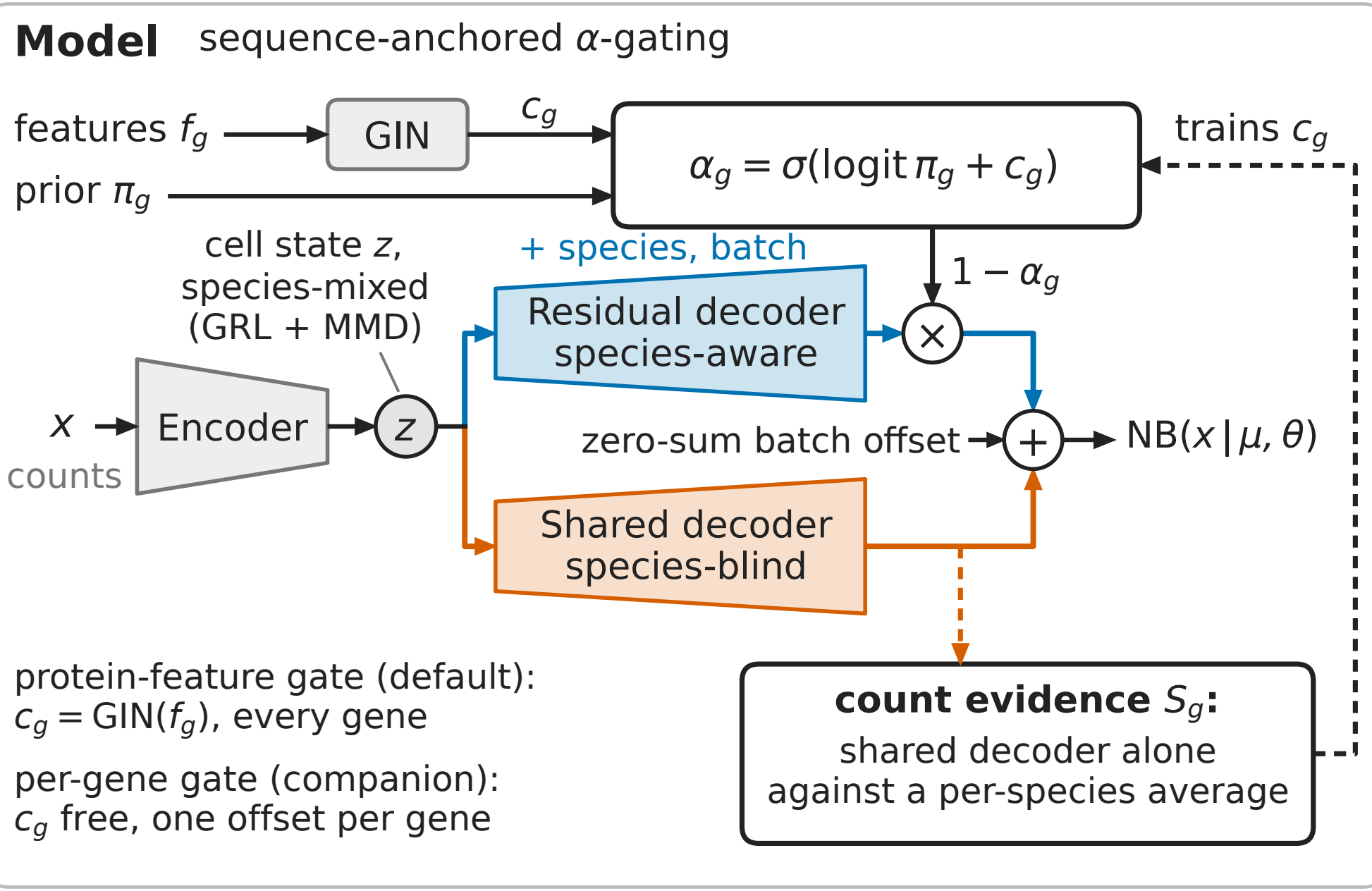
Model
sequence-anchored α-gating
features f_g
prior π_g
GIN
c_g
$\alpha_g = \sigma(\text{logit}\,\pi_g + c_g)$
trains c_g
cell state z,
species-mixed
(GRL + MMD)
+ species, batch
$1 - \alpha_g$
Residual decoder
species-aware
x
counts
Encoder
z
zero-sum batch offset
$\text{NB}(x \mid \mu, \theta)$
Shared decoder
species-blind
protein-feature gate (default):
$c_g = \text{GIN}(f_g)$, every gene
per-gene gate (companion):
c_g free, one offset per gene
count evidence S_g:
shared decoder alone
against a per-species average